

Machine Learning

Course Code	CSE 303	Course Category	CC		L	T	P	C
					3	0	1	4
Pre-Requisite Course(s)		Co-Requisite Course(s)		Progressive Course(s)				
Course Offering Department	CSE	Professional / Licensing Standards						

Course Objectives / Course Learning Rationales (CLRs)

1. Introduce Machine Learning and various tasks involved in the pipeline of machine learning application development.
2. Understand a wide variety of regression, classification and clustering algorithms.
3. Apply the algorithms to a real-world problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models.
4. Learn the rapid advances in Machine Learning and be able to understand the research articles.

Course Outcomes / Course Learning Outcomes (CLOs)

	At the end of the course the learner will be able to	Bloom's Level	Expected Proficiency Percentage	Expected Attainment Percentage
Outcome 1	Demonstrate the phases of machine learning application development.	2	75%	75%
Outcome 2	Describe the learning algorithms.	2	75%	70%
Outcome 3	Explain the techniques to deal with data and its dimension.	2	70%	65%
Outcome 4	Develop speech recognition, object recognition and classification models using machine learning algorithms	4	70%	65%

Course Articulation Matrix (CLO) to Program Learning Outcomes (PLO)

CLOs	Program Learning Outcomes (PLO)														
	Engineering Knowledge	Problem Analysis	Design and Development	Analysis, Design and Research	Modern Tool and CT Usage	Society and Multicultural Skills	Environment and Sustainability	Moral, and Ethical Awareness	Individual and Teamwork Skills	Communication Skills	Project Management and Finance	Self-Directed and Life Long Learning	PSO 1	PSO 2	PSO 3
Outcome 1	3	3	3	-	-	-	-	-	-	-	-	-	3	2	
Outcome 2	3	3	3		2	-	-	-	-	-	-	-	3	3	
Outcome 3	3	3	2	-	-	-	-	-	-	-	-	-	3	2	
Outcome 4	3	3	2	-	-	-	-	-	-	-	-	-	3	3	
Average	3	3	3		2								3	3	

Course Unitization Plan Theory

Unit No.	Unit Name	Required Contact Hours	CLOs Addressed	References Used
UNIT I		12		
	Introduction: Introduction to Machine Learning	1	1	1
	Different types of learning	1	1	1
	Different models and Learning algorithm	1	1	1
	Hypothesis space and inductive bias	1	1	1
	Training, Testing, validation of models	1	3	2
	Evaluation of the model: Train data, Test data	1	3	2
	Evaluation of the model: Cross Validation, Overfitting and Underfitting	1	3	2
	Regression: Introduction	1	2	3
	Linear Regression: Simple	1	2,4	3
	Linear Regression: Multiple	1	2,4	3
	Polynomial regression	1	2,4	3
	Evaluating regression fit	1	2,4	3
UNIT II		13		
	Decision tree learning: Introduction, Decision tree representation	1	2,4	1
	appropriate problems for decision tree learning, the basic decision tree algorithm	1	2,4	1
	hypothesis space search in decision tree learning, inductive bias in decision tree learning,	1	2,4	1
	issues in decision tree learning	1	2,4	1
	Decision tree learning (ID3) Algorithm and numerical	1	2,4	1
	Instance based Learning: K nearest neighbor, numerical problem	1	2,4	1
	the Curse of Dimensionality, Feature selection	1	2,4	1
	Univariate and Multivariate feature selection approaches	1	2,4	1
	Feature selection techniques	1	2,4	1
	Feature reduction: Principal Component Analysis	1	2,4	1
	Feature reduction: Principal Component Analysis	1	2,4	1
	Feature reduction: Linear Discriminant Analysis	1	2,4	1
	Recommender System: Content based system, Collaborative filtering based	1	2,4	4
UNIT III		4		
	Probability and Bayes Learning: Probability and classification, Bayesian Learning,	1	2	1
	Bayes optimal decisions, Naïve Bayes	1	2,4	1
	Support Vector Machine: Introduction, the Dual formulation,	1	2,4	1
	Maximum margin with noise, nonlinear SVM and Kernel function, solution to dual problem, python exercise on SVM	1	2,4	1
UNIT IV		11		
	Artificial Neural Networks: Introduction, , Biological motivation, ANN representation	1	2,4	2
	appropriate problem for ANN learning, McCulloch-Pitts neuron	1	2,4	2
	Perceptron, Perceptron learning, implementation of logic gates using perceptron	1	2,4	2
	Problem with perceptron, Gradient descent algorithm	1	2,4	2
	ADALINE and delta rule, implementation of logic gates using ADALINE	1	2,4	2
	Problem with ADALINE, Nonlinear classification using ADALINE: Polynomial discriminant function, MADALINE	1	2,4	2

	multilayer networks and the back propagation algorithm	1	2,4	2
	Radial Basis Function Neural Network	1	2,4	2
	Radial Basis Function Neural Network	1	2,4	2
	Introduction to Computational Learning Theory: Introduction	1	2	1
	sample complexity, finite hypothesis space, VC dimension	1	2	1
UNIT V		5		
	Ensembles: Introduction, Bagging and boosting, Random Forest	1	2,4	3
	Fixed rule fusion techniques, Trained rule fusion techniques	1	2,4	3
	Trained rule fusion techniques	1	2,4	3
	Clustering: Introduction, K-mean clustering	1	2,4	3
	Hierarchical clustering	1	2,4	3
	Total contact hours		45	

Course Utilization Plan- Lab

Exp. No.	Experiment Name	Required Contact Hours	CLOs Addressed	References Used
1	Introduction to Python basics	2	4	4
2	Machine Learning packages in Python	2	4	4
3	Implement different types of regression using python	2	4	4
4	Write a program that provides an option to compute different distance measures between two points in the N dimensional feature space. Consider some sample datasets for computing distances among sample points	2	4	4
5	Implement k-Nearest Neighbour algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.	2	4	4
6	Implement ID3 algorithm to construct a decision tree. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample	2	4	4
7	Given a dataset. Write a program to compute the Covariance, Correlation between a pair of attributes. Extend the program to compute the Covariance Matrix and Correlation Matrix	2	4	4
8	Write a program to implement feature reduction using Principal Component Analysis	2	4	4
9	Write a program to implement the naïve Bayesian classifier for a sample training data set. Compute the accuracy of the classifier, considering few test data sets.	2	4	4
10	Given a dataset for classification task. Write a program to implement Support Vector Machine and estimate its test performance.	2	4	4
11	Write a program to implement perceptron for different learning tasks.	2	4	2
12	Write programs to implement ADALINE and MADALINE for a given learning task.	2	4	2
13	Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets	2	4	2
14	Write a program to implement the K-means clustering algorithm. Select your own dataset to test the program. Demonstrate the nature of output with varying value of K	2	4	4
15	Implementation of hierarchical clustering using python	2	4	5
			30	

Learning Assessment Theory

Bloom's Level of Cognitive Task		Continuous Learning Assessments (30%)				End Semester Exam (30%)
		CLA-1 (5%)	Mid-1 (10%)	CLA-2 (5%)	CLA-3 (10%)	
Level 1	Remember	70%	50%	40%	40%	40%
	Understand					
Level 2	Apply	30%	50%	60%	60%	60%
	Analyse					
Level 3	Evaluate					
	Create					
Total		100%	100%	100%	100%	100%

Learning Assessment Lab

Bloom's Level of Cognitive Task		Lab Performance (20%)	End Semester Exam (20%)
Level 1	Remember	20%	30%
	Understand		
Level 2	Apply	80%	70%
	Analyse		
Level 3	Evaluate		
	Create		
Total		100%	100%

Recommended Resources

1. Mitchell, T.M. and Tom, M. (1997) Machine Learning. McGraw-Hill, New York.
2. Deepa, S. N., & Sivanandam, S. N. (2011). Principles of soft computing. Delhi, India: Wiley India Pvt. Ltd
3. Alpaydin, E. (2020). Introduction to machine learning. MIT press.
4. Swamynathan, M. (2017). Mastering machine learning with python in six steps: A practical implementation guide to predictive data analytics using python. Manohar Swamynathan.

Other Resources

1. Bishop, C. M., & Nasrabadi, N. M. (2006). Pattern recognition and machine learning (Vol. 4, No. 4, p. 738). New York: springer.

Course Designers